FHIR + AI Hackathon - Session 3 Orientation

Session 3: Open-Ended Agent Exploration

Duration: 1 hour

Your Goals Today

1. Run at least 2 different clinical questions through the agent

This is exploratory - there are no "right" questions!

What's New: 6 Tools Available

From Session 2 (3 tools):

New in Session 3 (3 tools):

The agent can now answer richer clinical questions!

Tool 4: search_medications

What it does:

Input:

Output:

Use cases: "What medications is this patient on?", "Are diabetic patients on insulin?"

Tool 5: search_encounters

What it does:

Input:

Output:

Use cases: "How many ER visits?", "When was their last appointment?"

Tool 6: search_all_conditions

What it does:

Input:

Output:

Use cases: "What comorbidities do diabetic patients have?", "Find patients with multiple chronic conditions"

Clinical Codes Reference

| Code | System | Meaning | Reference |

HbA1c Interpretation:

Question Ideas: Straightforward

These should work well:

1. *"Find patients with Type 2 diabetes and list their current medications"*

2. *"Find patients with hypertension (SNOMED CT: 59621000) and their most recent blood pressure (LOINC 85354-9)"*

3. *"How many encounters did diabetic patients have?"*

Question Ideas: More Complex

These require creative chaining:

4. *"Find patients with both diabetes and hypertension - what medications are they on?"*

5. *"Are there patients with diabetes who have had emergency encounters?"*

6. *"Find diabetic patients with declining kidney function (creatinine LOINC 2160-0)"*

Question Ideas: Edge Cases

These may cause the agent to struggle:

7. *"Which patients are the sickest?"*

8. *"Find patients at risk for diabetic complications"*

9. *"Compare treatment patterns across diabetic patients"*

Failure Mode 1: Hallucination

What it looks like:

Example:

Why it happens:

How to catch it:

Failure Mode 2: Wrong Tool Choice

What it looks like:

Example:

Why it happens:

How to catch it:

Failure Mode 3: Premature Termination

What it looks like:

Example:

Why it happens:

How to catch it:

Failure Mode 4: Over-Querying

What it looks like:

Example:

Why it happens:

How to catch it:

The Portability Problem

What we hard-coded in Sessions 2-3:

python

available_functions = {

Pain points:

What if we want to share these FHIR tools with other developers or AI apps?

Model Context Protocol (MCP)

What it solves:

MCP Architecture:

MCP Client (Claude Desktop, IDE, custom app)

Our Notebook vs MCP

Our Notebook:

MCP Architecture:

The shapes are the same. MCP standardizes the connections.

What Would Our FHIR Tools Look Like as an MCP Server?

Conceptual sketch (not code):

FHIR MCP Server exposes:

Any MCP client (Claude Desktop, IDE plugin, custom app) can:

You built the hard part - the tools. MCP is just the packaging.

MCP in Practice

Example: Claude Desktop

With an MCP-enabled FHIR server:

No notebook required!

MCP Resources

Official Docs:

MCP Servers:

Building Your Own:

Our hackathon showed you the concepts. MCP is the production implementation.

Your Deliverable Today

What to Submit:

File: hackathon_session3_{your_student_id}.json

Contents:

Generated automatically by the notebook at the end!

Reflection Questions (Required)

Answer in the notebook:

1. What surprised you most about how the agent handled your questions?

2. When would you trust an agent like this with real clinical data? What safeguards would you want?

3. How does tool use relate to MCP? Based on what you've seen, what does MCP standardize that our notebook left ad hoc?

These help you synthesize what you learned!

Tips for Success

1. Start simple, then increase complexity
2. Watch for all 4 failure modes
3. Check the trace carefully
4. Document surprises

What to Expect

[OK] Success:

[!] Common observations:

Remember: Finding failures is SUCCESS! That's the learning goal.

Questions?

Before you start:

Let's explore!

Key Takeaways

What You've Learned:

Why It Matters:

Next Steps:

Resources

- FHIR Server: https://launch.smarthealthit.org/v/r4/fhir

Key Codes:

Good luck!